Phase 1: Lightweight Spike-Based DOA Sector Inference for Embedded Edge AI

Hossein Molhem

Kennesaw State University
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Motivation

- Edge sensing systems require inference under compute, memory, energy, and latency constraints.
- Spike-based inference is attractive because it can represent information through sparse temporal events.
- DOA sector classification is a useful test case because it connects sensing physics, signal processing, and embedded AI.

Phase 1 Objective

Goal

Build a reproducible Python reference prototype for DOA-sector inference using a ULA signal model, beamspace features, rate-based spike encoding, and LIF-inspired temporal classification.

- Not a claim of a new DOA algorithm.
- Establishes a controlled engineering baseline.
- Creates a bridge toward PYNQ, FPGA, and Vitis HLS implementation.

End-to-End Pipeline

ULA / DOA signal model

$\Downarrow$

Array snapshots with controlled SNR

$\Downarrow$

Sample covariance matrix R_{xx}

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Beamspace features

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Feature normalization

$\Downarrow$

Nearest-centroid baseline and LIF-inspired classifier

Theoretical Foundation

$$a(\theta) = [1, e^{-jkd\sin\theta}, \ldots, e^{-j(N-1)kd\sin\theta}]^T$$

$$R_{xx} = \frac{1}{M} X X^H$$

$$p(\phi) = b^H(\phi) R_{xx} b(\phi)$$

$$\hat{c} = \arg\min_c \|x - \mu_c\|_2^2$$

$$v_c[t] = \alpha v_c[t-1] + i_c[t]$$

- ULA phase progression creates angle-dependent spatial structure.
- Beamspace energy converts covariance information into compact features.
- Spike encoding and LIF accumulation create the temporal edge-inference path.

Signal Model

For each source angle θ and SNR value:

$$X = a(\theta)s + N$$

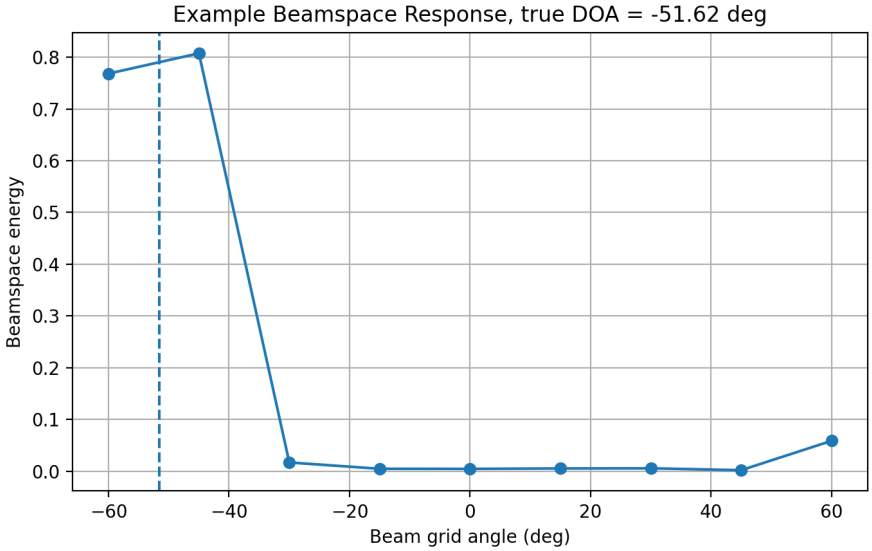
$$R_{xx} = \frac{1}{M} XX^H, \qquad M = 128$$

- $a(\theta)$: ULA steering vector.
- s: complex narrowband source signal.
- N: circular complex Gaussian noise.
- Beamspace energy features are extracted from R_{xx}.

Dataset Configuration

Parameter	Value
Array elements	8
Snapshots	128
Spacing	$\lambda/2$
DOA range	-60° to $+60^\circ$
Sectors	4
SNR values	$-10, -5, 0, 5, 10, 15, 20$ dB
Total samples	2800
Test samples	840

Beamspace Response Example



Baseline and LIF Inference

Nearest-centroid baseline

- Continuous normalized features.
- Deterministic and interpretable.
- Overall accuracy: 0.869048.

LIF-inspired classifier

- Rate-coded spike trains.
- Leaky temporal evidence accumulation.
- Fixed Run 003 accuracy: 0.825000.

LIF Temporal Accumulation

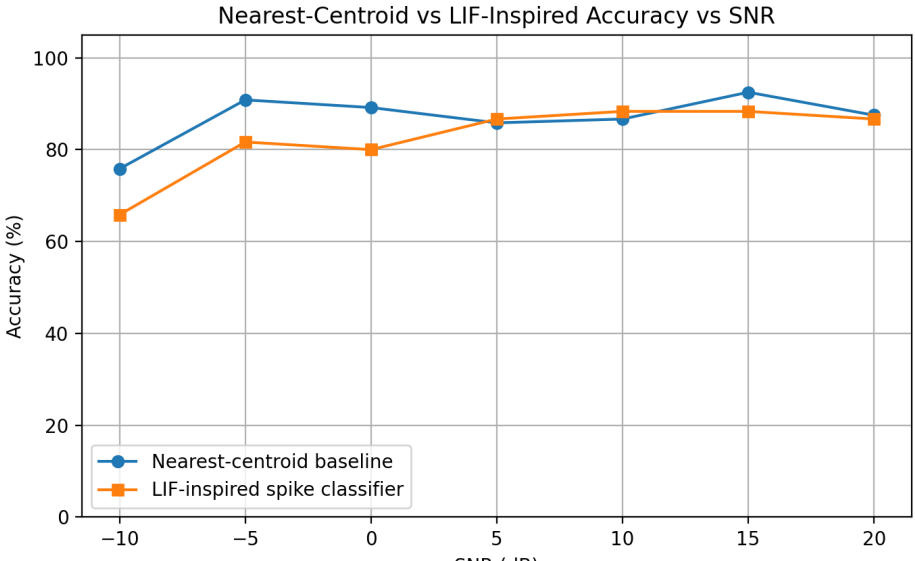
$$v_c[t] = \alpha v_c[t-1] + i_c[t]$$

- $v_c[t]$: membrane evidence for class c.
- α: leak factor.
- $i_c[t]$: spike-driven input current.
- Stable-decision latency is the earliest time after which the final decision remains unchanged.

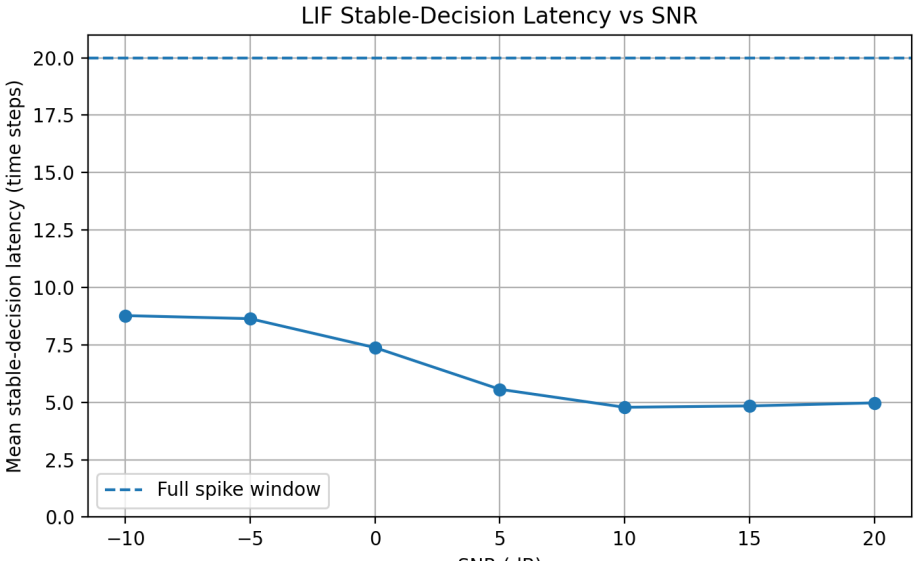
Fixed Run 003

$T = 20$, leak $= 0.97$, accuracy $= 0.825000$, mean latency $= 6.423810$ time steps.

Accuracy Comparison



Latency Versus SNR

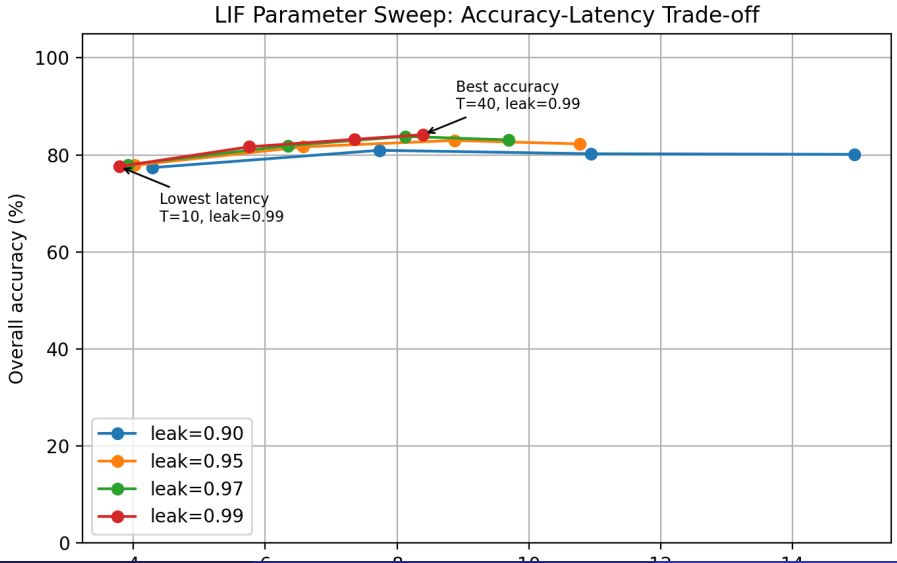


Run 005 Parameter Sweep

Case	T	Leak	Accuracy	Latency
Best LIF accuracy	40	0.99	0.841667	8.396429
Lowest latency	10	0.99	0.776190	3.782143
Fixed Run 003	20	0.97	0.825000	6.423810
Baseline	–	–	0.869048	–

The sweep exposes a direct accuracy–latency design trade-off.

Accuracy–Latency Trade-Off



Engineering Interpretation

- Beamspace features are informative for coarse DOA-sector inference.
- The spike path does not improve raw accuracy over the continuous baseline.
- Its value is architectural: sparse temporal inference, measurable latency, and tunable accuracy–latency behavior.
- LIF parameters provide explicit control through time-window length and leak factor.
- Phase 1 is a stable reference model for embedded implementation and hardware-aware refactoring.

Conclusion and Next Steps

Conclusion

Phase 1 establishes a reproducible DOA-sector inference baseline. The spike-based path trades some continuous-valued accuracy for sparse temporal processing and explicit latency control.

- Baseline accuracy: 86.90%; best LIF accuracy: 84.17%.
- Use the model as a hardware-aware embedded-AI reference path.
- Extend toward richer radar/DOA case studies and PYNQ or Vitis HLS refactoring.